



Introduction to Machine Learning





Definition:

Machine learning means teaching computers to recognize patterns in data, learn from those patterns, and make decisions or predictions based on new input data.



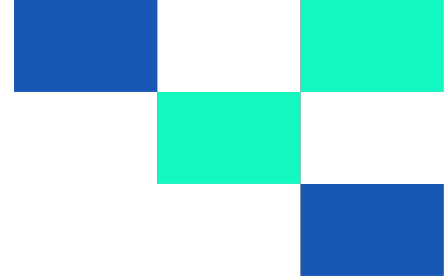


Key Idea

Instead of writing rules to detect spam emails, you give the ML model thousands of emails labeled as "spam" or "not spam," and it learns what characteristics make an email spam.



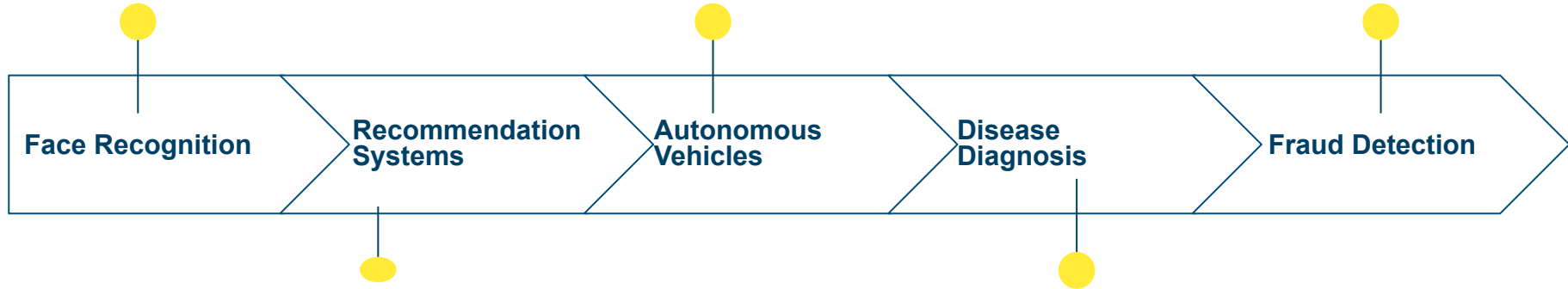
Applications of Machine Learning:



Used in mobile phones, security systems, and social media platforms.

Enables self-driving cars to recognize obstacles, follow traffic rules, and make driving decisions.

Identifies suspicious activity in banking and finance, such as unusual credit card transactions.



Face Recognition

Recommendation Systems

Autonomous Vehicles

Disease Diagnosis

Fraud Detection

Suggests products, movies, or music based on user behavior (e.g., YouTube, Netflix, Amazon).

Helps doctors detect diseases like cancer or diabetes using medical imaging and patient data.

Types of Machine Learning



Supervised Learning

The model learns from data that already has the correct answers (labels).

Example:

Predicting house price

- Inputs: size, location, rooms
- Output: price (known in training data)

Unsupervised Learning

The model finds patterns in data without knowing the correct answers.

Example:

Grouping customers based on spending behavior

Types of Supervised Learning



Regression

Regression is a type of supervised learning where a model is trained to predict a number.

It learns the relationship between input data and a continuous value, such as price, age, or temperature.

Classification

Classification is a type of supervised learning where a model learns from examples and then assigns items to categories or groups.

Just like separating fruits into apples and oranges, the computer uses data to decide which group something belongs to.



Regression Algorithms

- Linear Regression
- Multiple Linear
- Polynomial Regression
- Tree-Based Regression
 - Decision Tree Regression
 - Random Forest Regression
- XGBoost Regression

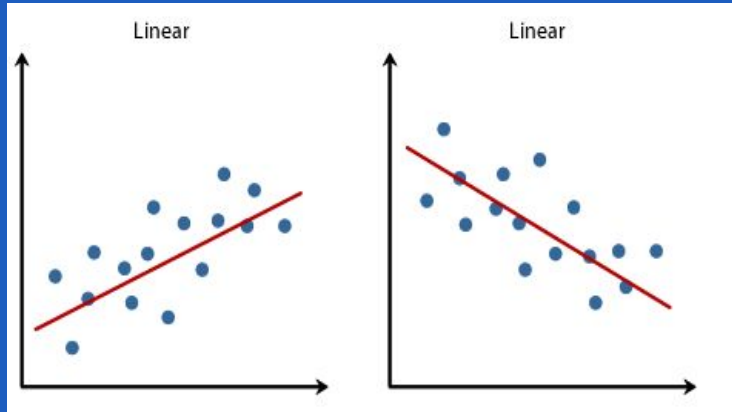
Classification Algorithm

- **Logistic Regression**
- **K-Nearest Neighbors (KNN)**

Linear Regression

Linear regression is a method used to predict numbers by assuming a straight-line relationship between input and output.

This means that as the input increases, the output increases or decreases at a steady rate.



Linear regression assumes there's a straight-line relationship between input features and the target variable

Example: Let's say you're studying how hours of study (X) affect exam score (Y).

If every extra 1 hour of study adds 10 points to the score, the pattern would be:

Hours (X)	Score (Y)
1	50
2	60
3	70
4	80

How Linear Regression Works

Linear regression finds the best-fitting straight line that describes the relationship between input (X) and output (Y). It uses that line to make predictions on new data.

Collect Data

You start with data that has:

- Input features (X) — e.g., hours studied
- Target values (Y) — e.g., exam scores

Plot the Data

On a graph:

- X-axis = hours studied
- Y-axis = exam score
- Plot each data point
- The goal is to find a line that best fits these points

The model tries to find the **line** that: $\hat{Y} = wX + b$

Where:

- $\hat{Y}$ is the predicted score
- X is the input (e.g., hours)
- w is the **slope** (how much Y changes when X changes)
- b is the **intercept** (where the line starts on the Y-axis)

Unsupervised Learning

In unsupervised learning, the algorithm is given unlabeled data, meaning there are inputs (X) but no corresponding output labels (Y). The goal is for the model to identify hidden patterns, structures, or groupings in the data without prior guidance.

The main categories of unsupervised learning include:

- **Clustering:**

Clustering involves grouping similar data points together based on certain features or patterns.

- **Association Rule Learning:**

Association rule learning is an unsupervised learning technique used to discover interesting relationships or patterns between variables in large datasets.



Comparison Table

Feature	Supervised	Unsupervised
Labels	Yes	No
Goal	Predict known outcomes	Discover hidden patterns
Example	House price prediction	Customer segmentation



THANK YOU